

# Deep Learning-Based Method for Irrigation Status Detection in Tomato Using Plant Leaves

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**Abstract**—The impact of climate change, arguably global warming and resulting drought, is one of the most escalating agricultural challenges affecting crop productivity. Therefore, effective water management is critical in agricultural practices. The analysis of plant leaves presents an opportunity to gauge irrigation status through automated solutions to encourage broader adoption among farmers. Currently, there is a notable absence of AI methods in the literature for detecting tomato plant irrigation status through leaf analysis. Addressing this gap, we propose a novel end-to-end deep learning (DL)-based method, inspired by the ResNet-50 model. Our model trims unnecessary blocks and reduces larger kernels, significantly streamlining the model to better fit with the leaf image dataset related to the tomato irrigation status. We evaluate our method using a newly developed dataset and find outstanding performance (Precision: 99.05%, Recall: 99.01%, F1-score: 99.01%, mean-average F1: 98.98%, weighted-average F1: 98.95%, Kappa: 98.61%, accuracy: 98.90%) while comparing with the pretrained DL models. Additionally, our model has fewer parameters and lower floating-point operations (FLOPs), enhancing its efficiency and suggesting its potential for more cost-effective and productive irrigation management practices.

**Impact Statement**—The proposed photograph-based leaf water stress detection technology provides the water quantity required in tomato plants. This establishes a foundation for the next step: developing the model as an application (either in mobile device or in server) for effective management of limited water resources. This application will enable users to check water

stress and apply precise amount of water either manually or through the automation of existing irrigation systems such as drip, sprinkler, subsurface, or other systems. This approach saves water while optimizing the productivity. Additionally, the technology is affordable and convenient to use for small scale farmers. The proposed technology has the potential to be applied to a range of crops and geographies, provided the model is adapted to fit the specific crops and topographies.

**Index Terms**—Artificial intelligence, convolution neural network, deep learning, irrigation status, precision agriculture, water management.

## I. INTRODUCTION

THE small-scale farmers, who cultivate less than 2 hectares of agricultural land, contribute significantly to global food production. The proportion of small-scale farmers is even larger in developing countries than in other parts of the world [1]. Most of them struggle with little or no access to information and practical solutions to optimize farming while facing new challenges to crop production such as insect outbursts, loss of nutrients in the soil, increased frequency of drought, drying up of springs, depletion of groundwater level, and continuous land degradation among others [2], [3].

To deal with such emerging challenges, farmers need precise information about their crops, so that they can make informed decisions and take precautions early through precision agriculture (PA) [4]. PA is a data-driven approach that tailors farming methods to individual plants or specific areas within a field, utilizing advanced technologies such as proximal sensors (e.g. soil sensors), Internet of Things (IoT), and geographical information system (GIS). Such practices can help the optimum uses of water used for irrigation, nutrient application, pest control, and other input resources, while increasing yields [5].

The availability of a wide range of field sensors and kits such as control systems, autonomous vehicles, drones and software made it possible to keep track of the nutrient and moisture availability on soils in very high spatial resolutions. Utilizing such information, the required quantity of irrigation, nutrients and pesticides can be recommended at the right time and place. However, the initial setup and operational cost of such sophisticated technologies are very high and not affordable to small-scale farmers [6]. The increasing trend of winter drought has added the irrigation management challenges to the farmers. Therefore, exploring cutting-edge technology that is affordable,

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easy to implement, and water-efficient is very much needed for small-scale farmers. This can help increase productivity in the winter and in nonirrigated mountainous regions where water is highly scarce. The proposed photograph of leaf-based irrigation management technology offers the quantity required for the particular plant of the tomato. This lays the foundation for the next step of deploying on mobile devices or integrate with the existing systems such as drip irrigation to automate and making precise uses of water for irrigation which is cost-effective and easy to use even for poor and less educated farmers.

Owing to such challenges, researchers have investigated developing a low-cost affordable system for automated plant phenotyping [4]. Plant phenotyping helps characterize the plant conditions such as colors, shape, and size over the crop life-cycle, which are the most critical indicators to detect the various types of stress [7]. The excess or inadequate use of water could have water-induced stress that hinders the photosynthesis process and growth of plants [8]. Therefore, better water management is crucial for products' quality and quantity. Additionally, farmers require an easy-to-use, reliable, and cost-effective solution for effective water management systems.

In the recent past, deep learning (DL)-based methods have been used extensively in automating plant phenotyping for the PA applications [9]. The images acquired from remote sensing technology or field-based sensors have been used commonly to characterize the plant's traits such as water deficiency monitoring [4], yield estimation [10], weed detection [11], growth monitoring [12], leaf disease detection [13], [14], and so on in various crops [4], [15]. Various sensors such as multispectral, hyperspectral, and RGB have been used for such monitoring [5]. For instance, thermal images were used to identify water deficits in sugarcane fields by de Melo et al. [16]. Among them, RGB sensors are the most convenient, cost-effective and most commonly used in the image analysis framework for crop health monitoring and disease detection using leaf images [17], [18]. For instance, Azimi et al. [4] implemented the DL-based methods for water stress classification on chickpea plants. They achieved the highest classification accuracy of 98.52% using VGG-16 [19] and InceptionV3 [20] as feature extractors on the entire plant shoot image dataset. A comparative assessment of three deep learning architectures, namely, AlexNet [21], GoogLeNet [22], and InceptionV3 [20] for water stress classification on three crops (maize, okra, and soybean) was investigated by Chandel et al. [23] using plant leaf images. The highest classification accuracy of 98.3% for maize, 97.5% for okra and 94.1% for soybean, was reported using GoogLeNet. Furthermore, a plant stress measurement due to nitrogen deficiency on the Sorghum plant was investigated by Azim et al. [24] using a 23-layered DL architecture. They compared the performance of a 23-layer DL architecture with the well-established large deep learning architectures such as ResNet [25] and NasNet-large [26] and showed that their proposal achieved comparable performance. Similarly, Zhuang et al. performed the irrigation status detection on maize plant using the leaf phenotype feature learned with various convolution neural networks (CNNs)

and reported an accuracy of 79.14% (for VGG16), 79.02 % (ResNet50), 78.86% (Xception) and 88.41% (custom CNNs).

The visual characteristics associated with irrigation stress include wilting, leaf curling, yellowing, and premature leaf drop in tomato plants regardless of their species and varieties. And existing research works demonstrated that DL-based methods such as VGG-16 are able to achieve promising performance in abiotic stress classification [4], [5], [23]. For instance, ResNet provided an accuracy of 84% for stress detection in Sorghum due to nitrogen deficiency [4]. However, these methods consist of large numbers of trainable parameters, requiring higher computational resources (both storage and training time). Therefore, it is a necessity to systematically explore the possibility of reduced-sized DL-based methods for water stress classification. Furthermore, there has been no such analysis performed in case of the tomato crop, a widely cultivated vegetable crop that is particularly vulnerable to water stress in the drought-prone middle mountain region of Nepal, in the Nepali context (to the best of our knowledge). To overcome the aforementioned limitations, we present the lightweight (reduced-sized) DL-based method for irrigation status detection through the classification of tomato plants using leaf images with extensive experimental works. Our proposal is data-centric, which is focused on the tomato leaf dataset as the existing models, which are mostly heavyweight and unsuitable for such a dataset. The efficacy of our proposal is evaluated and validated using the novel tomato leaf images dataset.

In summary, the main *contributions* of this article include a novel database preparation and leaf-based DL model development to detect water stress in tomato plant that enables farmers to automate the irrigation management system. The specific contributions are listed as below.

- 1) We create a new tomato leaf dataset with a variable rate of irrigation status. Our dataset, which is open access and the only available dataset to the best of our knowledge in the Nepali context, is useful to the research/industrial community to detect tomato water stress for further research.
- 2) We develop a novel deep learning (DL)-based method for identifying water stress in tomatoes. This helps to automate the irrigation status detection on tomato leaves effectively and efficiently. To the best of our knowledge, this is the first study carried out to address the irrigation status detection in tomatoes using plant leaves.
- 3) We perform a detailed comparative study, ablation study and computational analysis of the developed DL-based method. Further, we show the efficacy of the proposed method using varying train/test splits on the dataset. This not only establishes the efficacy of the proposed method for tomato leaf stress detection but also provides state-of-the-art results.
- 4) We demonstrate the explainability of the results obtained from the proposed DL-based method using the gradient-weighted class activation mapping (Grad-CAM) technique [27]. This establishes trust and confidence in our method to the agricultural community.

The remaining article is organized as follows. Section II discusses the materials and proposed method. Section III discusses and analyzes the results compared with the off-the-shelf deep learning models and finally, Section IV concludes the article with future works.

## II. MATERIALS AND METHOD

### A. Dataset

To achieve tomato irrigation detection with leaf images, the tomato plants are cultivated in a greenhouse setting and the images of tomato leaves are recorded. The tomato leaf images are categorized into four classes based on the irrigation treatment as defined in Table I and Fig. 1. These irrigation treatment trials are designed based on widely accepted guidelines such as scientific robustness, data sufficiency, ability to track variability, and ease of communicating results as recommended in [28], [29]. Further, the simple statistics of the tomato leaf image dataset are presented in Table I. The detailed activities involved in tomato leaf image dataset creation are discussed in the following sections.

1) *Tomato Crop Cultivation and Treatment*: The tomato crop management experiment is carried out on the premises of Western Regional College (WRC), Pokhara, Nepal, from April 2019 to October 2019, shown in Fig. 2. This location comes under a subtropical and humid climate. However, the practice is executed under the greenhouse environment setup where the temperature ranges from 17°C to 38°C and no adjustments are made to light conditions. The size of the greenhouse is 10 m × 25 m. The soil is sandy loam with slight acidity and had medium organic matter, phosphorous and nitrogen, and low potassium in the study area. A total of 90 tomato plants are subjected to four different irrigation conditions (Table I). The soil moisture content was measured daily using time domain reflectometry (TDR) sensor. Irrigation was applied as needed to maintain the recommended (Table I) moisture levels throughout the study period.

We conduct photo point monitoring, a method involving the capturing of multiple photographs of the plants/leaves under different treatment conditions over a specific timeframe. Additionally, we analyze the coloration of the leaves. This comprehensive analysis aids in determining the irrigation needs of tomato plants as part of our moisture/irrigation requirement treatment.

2) *Plant and Leaf Selection*: We categorize the plants based on the treatments they received, specifically variable-rate irrigation. In each treatment group, we regularly capture images of a representative plant that accurately depicts the status of the particular treatment. These images served as a visual reference for monitoring the progress of each treatment. Since, the position of leaves in the plant exhibits different levels of wilting, freshness and edge curling in under-irrigated conditions, it is necessary to ensure a typical representation of the plant while collecting the leaves images. For this, we captured three images from each plant: one of the older leaves (first-generation leaves) at the bottom of the plant, one from the majority of leaves in

TABLE I  
DETAIL OF TOMATO LEAF IMAGE DATASET FOR EACH IRRIGATION CLASS AND TREATMENT

| Class                 | Treatment                      | #Samples     |
|-----------------------|--------------------------------|--------------|
| Low irrigated (LI)    | 25% of recommended irrigation  | 5247         |
| Medium irrigated (MI) | 50% of recommended irrigation  | 5170         |
| Highly irrigated (HI) | 75% of recommended irrigation  | 5108         |
| Fully irrigated (FI)  | 100% of recommended irrigation | 5430         |
| <b>Total</b>          |                                | <b>20955</b> |

the middle part of the plant, and one from the youngest leaf that has already reached its full size.

3) *Camera Settings and Image Acquisition*: We use a Canon EOS 200D 24.2-megapixel camera with an EF-S 18–55 mm f/3.5–5.6 lens (Canon, Japan) to capture the photographs. The optimal camera settings (aperture: 5.6, ISO: 220, focal length: 55 mm, and shutter speed: 1/60) were determined at the beginning of the trial and maintained consistently throughout the data collection experiment. Photographs were taken twice a week, every Monday and Thursday morning from 9:00 AM to 10:00 AM Nepal Standard Time (GMT+4:45). The following factors are considered when taking photographs repeatedly to maintain consistent conditions as closely as possible.

- A full leaf is covered in the image and the targeted leaf covers at least 70% of the photograph. If that is not possible, we crop the edge after taking the photo.
- The distance between the camera and the leaf kept same (~25 cm) all the time. We take the images from at least five directions for each leaf. However, we try to keep the same vertical and horizontal distance from the leaf all the time. We control the focal length to keep the same focal length all the time.

4) *Image Preprocessing*: We tag each image to one of the four categories (LI, MI, HI, and FI) to make sure that an image belongs to a specific tomato cultivation treatment category. Then, we remove the background of all images to reduce the background noise while training the models, perform the basic geometric transformations such as flip, crop, and rotation of the images for data augmentation during the training of the model [30] and crop all images to 224 by 224 pixels with 96 dpi to reduce the computing resources. Furthermore, the image pixels are normalised in the range of 0–1 for training the proposed model [25].

### B. Evaluation Metrics

We utilize seven metrics to measure and analyze the performance of the proposed DL-based method. These metrics include precision, recall, f1-score, accuracy, kappa-score, macroaverage f1-score (MAF1), and weighted-average f1-score (WAF1).

$$P = \frac{TP}{TP + FP} \quad (1)$$

$$R = \frac{TP}{TP + FN} \quad (2)$$

$$F1 = 2 \times \frac{P \times R}{P + R} \quad (3)$$

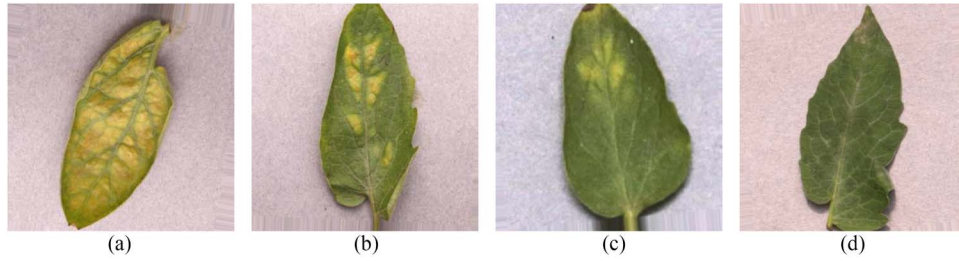


Fig. 1. Sample leaf images from tomato irrigation leaf dataset. Note that (a)–(d) denote low, medium, highly, and fully irrigated tomato leaves, respectively.



Fig. 2. Study site for tomato leaf image collection: (a) inside layout of the greenhouse; (b) emergence status of the tomato plant; (c) closeup shot of fully grown tomato plants; and (d) tomato plant ready for image acquisitions.

$$\text{Acc.} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (4)$$

where TP, TN, FP, and FN represent true positive, true negative, false positive, and false negative, respectively. Similarly,  $P$ ,  $R$ ,  $F1$ , and  $\text{Acc}$  represent Precision, Recall, F1-score, and Accuracy, respectively

$$\text{Kappa} = \frac{r_o - r_e}{1 - r_e} \quad (5)$$

where  $r_o$  is the observed agreement ratio and  $r_e$  is the hypothetical probability of chance agreement. MAF1 and WAF1 are derived from f1-score (F1) [31].

### C. Pretrained DL Models

Image classification has progressed well in recent decades because of the availability of various DL models trained on large datasets such as “ImageNet” [32]. For domain-specific image classification such as fruits images [31] and biomedical images [33], the transfer-learning approach [34] has been used along with the pretrained DL models. In transfer learning, basically, we take an architecture already trained on a large dataset and transfer the learned knowledge or feature to the small target dataset [35]. It can be achieved by fine-tuning the DL model with or without freezing the layers.

In the first approach, the pretrained DL model is again re-trained on the target dataset from scratch, which updates the weights of all layers of the DL model while training on the target dataset. It requires a relatively large dataset to optimally learn or reach the global optimum weights while training, otherwise there is always a risk of the DL model ending up in the local optimum value. The second approach involves freezing all the layers down to the classification layer and fine-tuning the top classification layer with a specific dataset. Freezing all the bottom layers in a DL model might miss the specific feature that

helps distinguish the irrigation status in the tomato leaf image dataset. Therefore, in this work, we employ the first approach, where we extract the feature map with each pretrained layer and add the two extra layers: global pooling layer (GAP) and the Softmax layer.

While looking into the literature, we found that both heavy-weight models such as VGG-16, ResNet, Inception [4] and light-weight models such as NASNet [24] were successfully employed for crop stress monitoring. While choosing pretrained DL models as contenders, we try to make the balance of both heavy-weight and light-weight models. Based on their uses in the water stress detection and numbers of parameters (i.e. weights), we ended up selecting ten pretrained DL models, which include both heavy-weight models such as InceptionV3 [20], VGG-16 [19], and Xception [36] and light-weight models such as NASNet, MobileNet [37], and EfficientNet [38] (refer to Table III). The results from these transfer learning approaches with fine-tuning are used to compare with the outcomes of the proposed method.

### D. Proposed Method

Given the efficacy of ResNet-50 on this dataset as seen in Table III, we design a novel DL-based method using it in this study. The ResNet-50 model comprises residual and skip connections, based on which they are able to capture more semantic information of the input dataset. The proposed ResNet-50 model is an end-to-end model, it automatically extract the features from the deep layers and classify using its SoftMax layer. It consists of residual connections but with significant modifications to achieve smaller layers, units and parameters without compromising the classification performance (Fig. 3).

Our DL-based method has a lower number of total parameters and floating point operations (FLOPs) (refer to Table VI) yet captures the highly discriminating information. This suggests

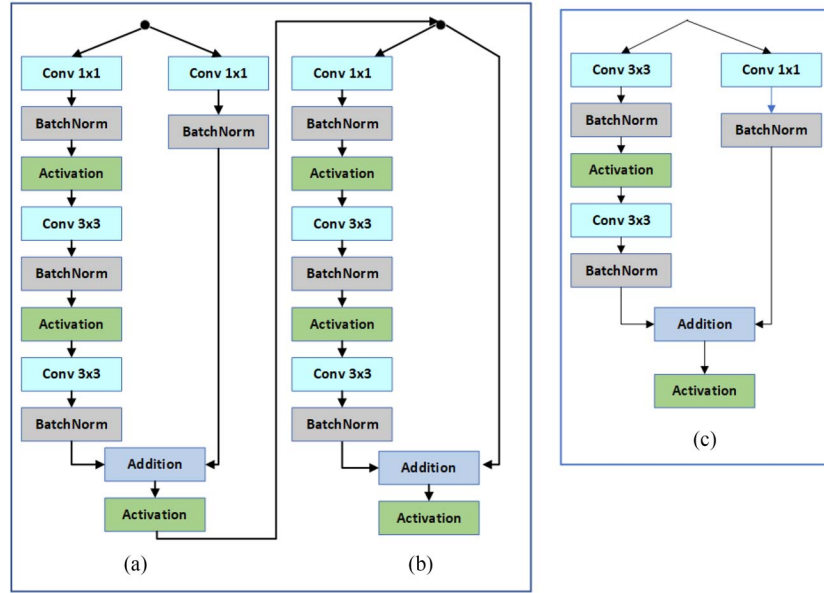


Fig. 3. Difference between original ResNet-50-based residual blocks (a)–(c), its modified lightweight version used in our work.

 TABLE II  
OVERALL ARCHITECTURE DETAILS OF OUR PROPOSED MODEL

| Block        | C_Layer | Operations     | P_layer | $(F, K, S)$ | Output       |
|--------------|---------|----------------|---------|-------------|--------------|
| Input block  | I1      | Input          | -       | -           | (224,224,3)  |
| Block-A      | A1      | Conv2D+BN+Relu | I1      | (16,3,2)    | (112,112,16) |
|              | A2      | Conv2D+BN+Relu | A1      | (16,3,2)    | (56,56,16)   |
|              | A3      | MaxPool2D      | A2      | -           | (28,28,16)   |
| Block-B      | B1      | Conv2D+BN+Relu | A3      | (32,3,1)    | (28,28,32)   |
|              | B2      | Conv2D+BN+Relu | B1      | (32,3,1)    | (28,28,32)   |
|              | B3      | Conv2D+BN+Relu | A3      | (32,1,1)    | (28,28,32)   |
|              | B4      | Add            | B2,B3   | -           | (28,28,32)   |
|              | B5      | Max-Pool2D     | B4      | -           | (14,14,32)   |
| Block-C      | C1      | Conv2D+BN+Relu | B5      | (64,3,1)    | (14,14,64)   |
|              | C2      | Conv2D+BN+Relu | C1      | (64,3,1)    | (14,14,64)   |
|              | C3      | Conv2D+BN+Relu | B5      | (64,1,1)    | (14,14,64)   |
|              | C4      | Add            | C2,C3   | -           | (14,14,64)   |
|              | C5      | MaxPool2D      | C4      | -           | (7,7,64)     |
| Block-D      | D1      | Conv2D+BN+Relu | C5      | (128,3,1)   | (7,7,128)    |
|              | D2      | Conv2D+BN+Relu | D1      | (128,3,1)   | (7,7,128)    |
|              | D3      | Conv2D+BN+Relu | C5      | (128,1,1)   | (7,7,128)    |
|              | D4      | Add            | D2,D3   | -           | (7,7,128)    |
|              | D5      | MaxPool2D      | D4      | -           | (3,3,128)    |
| Block-E      | E1      | Conv2D+BN+Relu | D5      | (256,3,1)   | (3,3,256)    |
|              | E2      | Conv2D+BN+Relu | E1      | (256,3,1)   | (3,3,256)    |
|              | E3      | Conv2D+BN+Relu | D5      | (256,1,1)   | (3,3,256)    |
|              | E4      | Add            | E2,E3   | -           | (3,3,256)    |
|              | E5      | MaxPool2D      | E4      | -           | (1,1,256)    |
| Output block | F1      | Flatten        | E4      | -           | 256          |
|              | F2      | Dense          | F1      | -           | 64           |
|              | F3      | softmax        | F2      | -           | 4            |

Note: F, K, and S denote filters, kernels, and strides used in each convolution operation, respectively. The C\_Layer and P\_Layer denote the current and previous layers in the CNN, respectively.

that the proposed model is more efficient and potentially less complex compared to the existing DL models. The overall architecture of our proposed method, containing seven different steps, is presented in Table II. Note that the proposed method

is an end-to-end DL model that automatically extracts features without requiring separate steps for feature extraction. It comprises seven different steps, which are explained in the following sections.

1) *Input Block (I)*: The input block takes a three-channel (RGB) leaf image as input with the size of  $(224 \times 224)$ . So, the original input images are resized accordingly while inputting into the CNN model. The output of this block is passed to the next block for processing.

2) *Block-A*: This is the first block of our proposed CNN, which consists of three types of layers: convolution layer, batch normalisation layer, and max-pooling. It includes two convolution layers, each layer having 16 filters and  $3 \times 3$  kernels with a stride of 2. These convolution layers are supplemented with batch normalisation and Relu activation functions. Finally, the max-pooling layer is used to reduce the feature map dimension before passing it to the next block.

3) *Block-B*: It is the first residual block in our CNN consisting of three convolution layers, each equipped with batch normalisation and Relu activation function. Here, in this block, we use 32 filters,  $3 \times 3$  size kernels with a stride of 1 for convolution operation. Two of the convolution layers are added in sequence, whereas the third convolution layer act as the residual connection to the previous block. The max-pooling operation is applied to reduce the dimension of the feature map before passing it to the next block.

4) *Block-C*: It also consists of three convolution layers with 64 filters,  $3 \times 3$  sized kernels, and a single stride. Here, the arrangement of these layers follows a similar fashion as in Block B.

5) *Block-D*: Block-D also follows a similar pattern as Block-C. However, the number of filters used in each convolution layer is increased to 128 in this block.

6) *Block-E*: Block-E also has the same number of layers as the previous three blocks. However, the number of filters used in each convolution layer is increased to 256 in this block.

7) *Output Block*: This is the final block in our architecture, which is used as a classification block, consisting of three layers: flatten, dense, and softmax. The dense layer has 64 neurons, whereas the softmax layer has four neurons to represent four tomato irrigation status classes.

### E. Implementation

We implement our research in Keras which has been executed using Python programming language [39]. We utilize the NVIDIA GeForce RTX 3070 Ti GPU with 32GB RAM. We split the tomato leaf images dataset into the train and test sets with a ratio of 70:30 per class. Five-fold cross-validation with random train/test splits was used to report the final averaged performance. To prevent the model from over-fitting during training, we set 10% of the training data for validation. We also utilise an Early stopping callback with the patience of three epochs with the validation loss during regularization. The other hyperparameters used for model training are set as a learning rate of 0.0001, Adam as an optimizer, categorical cross-entropy as loss function and batch size of 64. Furthermore, we apply the basic geometric transformations such as flip, crop, and rotation of the images for data augmentation during the training of the model [30]. The maximum epoch is set to 40. The best-fit curve for accuracy and loss metrics during training

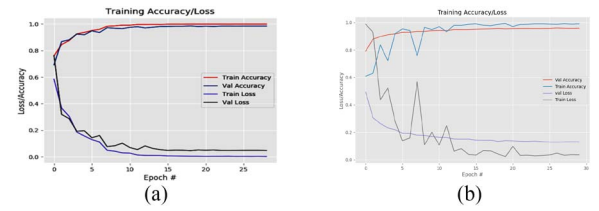


Fig. 4. Accuracy and loss curves for both training and validation of our model on Fold-1. (a) Proposed model. (b) ResNet50

and validation of our model is shown in Fig. 4(a), alongside that of ResNet50 Fig. 4(b). With these experimental settings, the proposed DL model is trained from scratch whereas the pretrained DL models (section II-C) were fine-tuned using a transfer learning approach. The results from these models were taken as reference while developing the proposed DL model.

## III. RESULTS AND DISCUSSION

Here, we compare the proposed method with the off-the-shelf pretrained DL models based on the standard evaluation measures. The results are presented in Table III. From Table III, we observe that our model imparts 99.05% of precision, 99.01% of recall and 99.01% of F1-score. This shows that our method delivers 7.70% higher in precision, 7.62% higher in recall and 7.76% higher in F1-score than the second-best performing method (ResNet-50). Also, our method imparts 30.24%, 32.9%, and 31.76% higher precision, recall, and F-score, respectively than the least-performing method (MobileNetV2). Furthermore, the features extracted from the pretrained DL models are based on training samples from the ImageNet dataset, which are more general in nature compared to the tomato leaf images used in this study.

Furthermore, Table III shows that our method provides 99.98%, 98.95%, 98.61%, and 98.90% for MAF1, WAF1, Kappa, and accuracy, respectively. This also highlights that our method exhibits 7.73% higher in MAF1, 7.60% higher in WAF1, 10.10% higher in Kappa, and 6.97% higher in accuracy than the second-best method (ResNet-50). From this result, we believe that the prominent results of our method in comparison to pretrained models are attributed to the smaller kernel size used in our residual blocks, which are able to extract more discriminating features at a deeper level from the tomato leaf images.

### A. Class-Wise Study

Here, the efficacy of our proposed method is also studied using class-wise performance measures, which are reported in Table IV. From Table IV, we observe that our method is able to produce a higher performance on the “low irrigated” class in terms of precision, recall, and f1-score, whereas it produces a comparatively lower performance for “medium irrigated” class. The reason for higher performance in the “low irrigated” class is the high defoliation of the tomato leaf due to water deficiency, whereas the degradation of performance in the “medium irrigated” class is due to a similar level of defoliation in medium

TABLE III  
COMPARATIVE STUDY OF PRETRAINED DL MODELS WITH THE PROPOSED METHOD USING  
PRECISION, RECALL, F1-SCORE, MAF1, WAF1, KAPPA-SCORE AND ACCURACY

| Model               | P            | R            | F1           | MAF1         | WAF1         | Kappa        | Acc.         |
|---------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| MobileNet [37]      | 82.17        | 82.52        | 82.25        | 82.06        | 82.24        | 76.68        | 82.52        |
| MobileNetV2 [40]    | 68.81        | 66.11        | 67.25        | 69.21        | 69.80        | 60.66        | 70.52        |
| NASNet-Mobile [26]  | 69.73        | 70.11        | 69.82        | 69.68        | 69.88        | 60.12        | 70.10        |
| EfficientNetB0 [38] | 88.41        | 88.51        | 88.41        | 88.28        | 88.40        | 84.67        | 88.50        |
| EfficientNetB1 [38] | 87.74        | 87.77        | 87.70        | 87.56        | 87.70        | 83.69        | 87.77        |
| DenseNet-121 [41]   | 83.89        | 84.07        | 83.89        | 83.72        | 83.88        | 78.75        | 84.07        |
| VGG-16 [19]         | 84.90        | 84.99        | 84.92        | 84.75        | 84.92        | 79.97        | 84.98        |
| ResNet-50 [25]      | <u>91.35</u> | <u>91.39</u> | <u>91.35</u> | <u>91.25</u> | <u>91.35</u> | <u>88.51</u> | <u>91.93</u> |
| Xception [36]       | 70.54        | 70.26        | 70.29        | 70.05        | 70.29        | 60.34        | 70.25        |
| InceptionV3 [20]    | 73.01        | 73.01        | 72.90        | 72.67        | 72.9         | 64.01        | 73.01        |
| <b>Ours</b>         | <b>99.05</b> | <b>99.01</b> | <b>99.01</b> | <b>98.98</b> | <b>98.95</b> | <b>98.61</b> | <b>98.90</b> |

Note: Boldface represents the highest performance, whereas underlined numbers denote the second-best performance.

TABLE IV  
AVERAGED CLASS-WISE PRECISION, RECALL,  
AND F1-SCORE OF OUR PROPOSED CNN ON FIVE  
TEST SPLITS (%)

| Classes | P      | R      | F1     |
|---------|--------|--------|--------|
| LI      | 100.00 | 100.00 | 100.00 |
| MI      | 97.25  | 98.65  | 98.10  |
| HI      | 98.35  | 97.25  | 98.45  |
| FI      | 99.20  | 99.50  | 99.25  |

TABLE V  
STUDY OF DIFFERENT RESIDUAL BLOCKS

| Model   | P     | R     | F1    | Acc.  |
|---------|-------|-------|-------|-------|
| $CNN_1$ | 92.10 | 92.31 | 92.01 | 92.22 |
| $CNN_2$ | 97.50 | 97.50 | 97.50 | 97.37 |
| $CNN_3$ | 98.75 | 98.75 | 98.75 | 98.48 |
| $CNN_4$ | 98.00 | 98.00 | 98.00 | 98.45 |
| $CNN_5$ | 99.00 | 99.00 | 99.00 | 98.90 |

|           |      |      |      |      |
|-----------|------|------|------|------|
|           | LI   | MI   | HI   | FI   |
| Actual LI | 1625 | 0    | 0    | 5    |
| Actual MI | 0    | 1553 | 22   | 0    |
| Actual HI | 0    | 32   | 1515 | 5    |
| Actual FI | 0    | 0    | 1    | 1532 |
|           | LI   | MI   | HI   | FI   |

Fig. 5. Confusion matrix of classification results for our model on Fold-1.

irrigated and highly irrigated leaf images. From this result, we believe that our proposed method works better to extract more discriminating information, representing the low irrigated class for high defoliation regions of the leave than other classes.

To further observe and analyze the performance at a deeper level, we present the confusion matrix representing all four classes for the testing split on Fold-1 in Fig. 5. From the figure, we notice that our method is able to separate the classes prominently. However, it mainly struggles between MI (midium irrigated) and HI (highly irrigated) classes

### B. Block-Wise Study

As a part of the ablation study, we find the contributing factors of our proposed method to improve performance. For this, we analyze each of the five different blocks (Block-A, Block-B, Block-C, Block-D, and Block-E) in terms of performance measures. Here, we design the following five CNNs using such blocks, where the input block and output block are common to all CNNs and each CNN is designed by adding

Block-A, Block-B and so on in sequential order. We denote these CNNs as  $CNN_1$  (with Residual Block A),  $CNN_2$  (With Residual Blocks A and B),  $CNN_3$  (With Residual Blocks A to C),  $CNN_4$  (With Residual Blocks A to D) and  $CNN_5$  (with Residual Blocks A to E) (Table V). To run individual CNNs, we add input and output block layers and train the model to report results. The results are presented in Table V. The result shows that the performance of the proposed method has increased from 92.22% accuracy (for  $CNN_1$ ) to 98.90% (for  $CNN_5$ ). The  $CNN_5$  containing all five blocks provides the highest accuracy in this study. However, the performance begins saturating from  $CNN_3$ , which contains three Blocks: Block-A, Block-B and Block-C. Furthermore, we carry out two more experiments with  $CNN_4$  and  $CNN_5$  with extra two blocks: Block-D and Block-E, respectively to see how much more performance can be achieved with such additions. This result, however, unveils that the performance reaches the peak on  $CNN_5$  with such five blocks. Therefore, we utilise five blocks in our proposed model.

### C. Qualitative Analysis

For the qualitative analysis, we leverage the visual activation maps of outputs achieved from our model using Grad-CAM [27]. The Grad-CAM is a model-specific technique that produces the class activation map highlighting the important regions or patterns in the image used for predicting the class. Here, we select a tomato leaf image for each irrigation class and find the activation maps, which are shown in Fig. 6. From the figure, we notice that the proposed method is able to learn the discriminative regions for each class as it highlights the yellow regions affected by low irrigation in each leaf image.

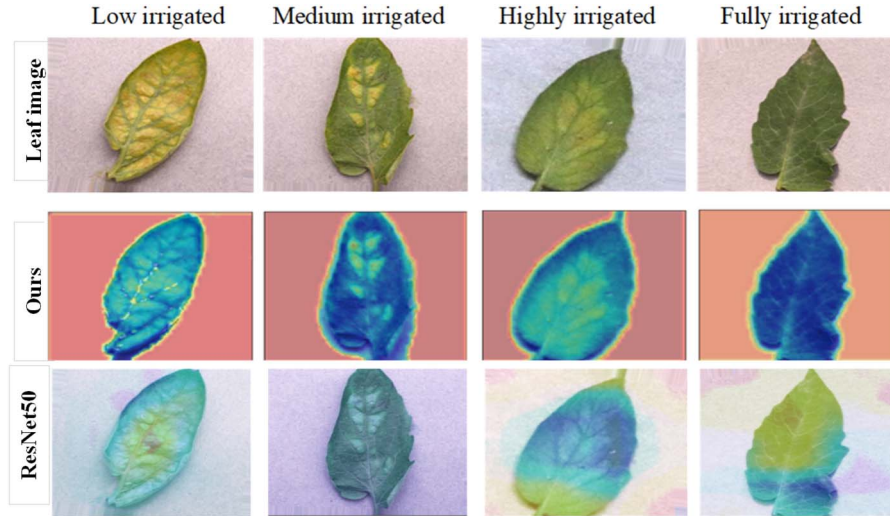


Fig. 6. Grad-CAM visualisation produced by ResNet50 and our model for four different classes.

For instance, the low irrigated image (first column in Fig. 6) has yellow regions over the majority area of the leaf, which has been captured by our proposed method. Similarly, the leaf image of the fully irrigated class (last row in Fig. 6) has all green regions and no yellow regions inside the leaf, which has been perfectly visualized in the corresponding Grad-CAM image as well. From this technique, we believe that the proposed method is able to correctly identify the deteriorated region in the leaf due to the low irrigation stress.

#### D. Computational Complexity Analysis

We utilize the floating point operations (FLOPs) to calculate and compare the computational complexity of the proposed model with the contender DL models used in this study. The FLOPs are computed for addition, subtraction, multiplication, division and any other operation that involves floating point value and are used to estimate the model's inference complexity. For instance, the FLOP operations for convolution and max-pooling layers are measured as  $(2 \times \text{filters} \times \text{KernelSize} \times \text{OutputShape})$  and  $(2 \times \text{InputSize} \times \text{OutputSize})$ , respectively. To calculate the total FLOPs for a particular CNN, the FLOPs of all layers are summed up and reported as total GFLOPs ( $\text{GFLOPs} = 10^9 \times \text{FLOPs}$ ). The comparison of the proposed model and the contenders in terms of computational complexity (trainable parameters and GFLOPs) is presented in Table VI. This shows the lower parameters and GFLOPs for our proposed model than contender DL models. Note that we utilise a Python package [42] that stands on the shoulders of the TensorFlow multistep profiler for the calculation of GFLOPs in our study.

#### E. Performance Analysis Over Multifolds Setting

We perform the DL models' performance analysis including ours in terms of accuracy on the tomato leaf image dataset. For this, we first design nine different random train/test fold settings: 1/9, 2/8, 3/7, 4/6, 5/5, 6/4, 7/3, 8/2, and 9/1. Then, we summarize all nine results for each DL model including ours

TABLE VI  
COMPARATIVE STUDY OF PRETRAINED DL MODELS  
WITH OURS USING PARAMETERS AND GFLOPs

| Model          | Param.      | GFLOPs      |
|----------------|-------------|-------------|
| MobileNet      | 3.23        | 1.14        |
| MobileNetV2    | 2.26        | 0.59        |
| NASNetMobile   | 4.23        | 1.13        |
| EfficientNetB0 | 4.01        | 0.78        |
| EfficientNetB1 | 6.51        | 1.16        |
| DenseNet-21    | 6.95        | 5.67        |
| VGG-16         | 14.71       | 30.70       |
| ResNet-50      | 23.59       | 7.72        |
| Xception       | 20.81       | 9.11        |
| InceptionV3    | 21.77       | 5.68        |
| <b>Ours</b>    | <b>1.24</b> | <b>0.11</b> |

Note: Param. denotes the trainable parameters in millions.

and present via a boxplot diagram (Fig. 7). From this figure, we find that our proposal imparts a consistent and higher performance while aggregating results over all the settings compared to other DL models.

#### F. Statistical Analysis Over Multifolds Setting

Given the importance of the statistical significance over performance measures obtained from eleven different DL models (including ours) as in Section III-E, we first test the normality and then perform the pairwise  $t$ -test. Here, we find the normal distribution of the results produced by our proposal and, therefore perform the parametric test (pairwise  $t$ -test) between ours and others. While performing pairwise  $t$ -test statistical analysis of our proposal with MobileNet, MobileNetV2, NASNetMobile, EfficientNetB0, EfficientNetB1, DenseNet-121, VGG-16, ResNet-50, Xception, and InceptionV3, it provides the  $p$ -value of 3.236e-13, 2.145e-15, 3.802e-16, 5.674e-10, 6.124e-12, 5.196e-08, 8.362e-08, 7.49e-13, 6.815e-10, and 8.988e-05, respectively. The results ( $p$ -value) show that they are extremely significant ( $p < 0.05$ ) during the pairwise  $t$ -test

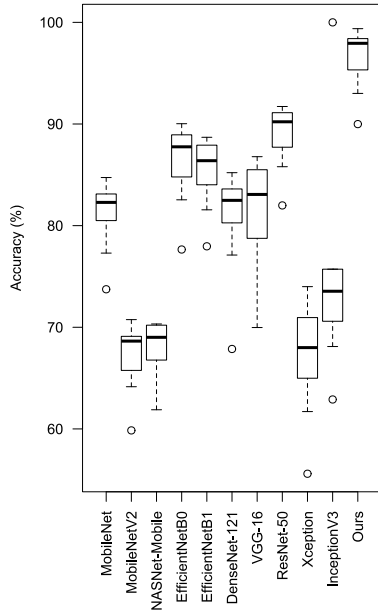


Fig. 7. Performance analysis (accuracy) across multifold settings for each DL model (including ours) and aggregation using boxplot.

with others. Therefore, our model provides more significant performance across accuracy measures, imparting robustness and generalizability compared to the contender methods as presented in Fig. 7. Furthermore, the statistical significance shows that the results are not randomized, meaning the classification accuracies are consistent across all the folds on the used dataset. As a result, this demonstrates the generalizability and robustness of the proposed model for tomato leave water stress detection.

The effects of irrigation deficiency on plants include wilting, leaf curling, yellowing, premature leaf drop etc. This study demonstrates that the effects of insufficient irrigation on tomatoes can be effectively detected, which is the main objective of this study. Considering all performance measures we considered, it is now clear that the proposed model outperforms the existing complex ResNet model, particularly for tomato leaves under different irrigation treatments. The proposed data-driven method has been customized to extract high-level semantic features that improve classification performance. In contrast, the existing ResNet model contains unnecessary layers that produce inefficient features, leading to higher complexity and yield relatively lower performance. Owing to its robustness, the proposed model has broader potential for application in diverse contexts, regardless of geography or crop type, as symptoms like wilting, leaf curling, yellowing, and premature leaf drop are common across different crop species across the world during irrigation deficiency. However, to achieve optimal performance, we recommend using transfer learning on other crop datasets with minimal effort instead of direct uses of this model. There is no need to train from scratch, as most data patterns in crop leaves are similar, as mentioned earlier. Additionally, it demonstrates the feasibility of developing such models and lays the foundation for the next step in our work: creating

lightweight models that can seamlessly integrate with mobile applications. This will provide farmers with an accessible and user-friendly tool to harness cutting-edge technology and maximize tomato yields and help farmers conserve resources and optimize their efforts. Additionally, as well as extend similar applications to other crops.

#### IV. CONCLUSION AND FUTURE WORKS

This article introduces a novel DL-based method that utilizes a modified version of the ResNet-50 architecture for efficient detection of irrigation status (specifically water deficiency) for improved water management in tomatoes and then validate via a detailed study on the newly collected tomato plant leaf image dataset. Our method achieves an accuracy of 98.80% for tomato irrigation detection. This result significantly outperforms the well-established DL models in terms of commonly used evaluation metrics, including precision, recall, f1-score, and so on. We also employ a Grad-CAM visualization of the output generated by our method to correctly identify the deteriorated regions on the leaf due to the low irrigation. This exhibits the explainability and interpretability of the proposed method. The findings demonstrate the possibility of the proposed method for real-time applications like plant stress monitoring and intelligent irrigation management. However, this method has two main limitations. First, the proposed method is focused only on the tomato leaf images dataset, so it would be interesting to see the performance of the method with more diverse datasets, for example, leaf images from different crops and environments. Second, the positional information of the leaf in the plant is not considered while training the models, so the effect of such information on irrigation status detection could be a future research work.

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#### CONFLICT OF INTEREST

The authors have no competing interests to declare that are relevant to the content of this article.

#### DATA AVAILABILITY

The datasets generated and analyzed during the current study are available at [43].

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